Neural Computation -- 1992

69 article(s), all issues

1. Learning curves for error minimum and maximum likelihood algorithms

KabashimaY., ShinomotoS.

(no DOI)

For the problem of dividing the space originally partitioned by a blurred boundary, every learning algorithm can make the probability of incorrect prediction of an individual example decrease with ...

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2. Neural Networks and the Bias/Variance Dilemma

Stuart Geman, Elie Bienenstock, René Doursat

https://doi.org/10.1162/neco.1992.4.1.1 | pp. 1-58

Feedforward neural networks trained by error backpropagation are examples of nonparametric regression estimators. We present a tutorial on nonparametric inference and its relation to neural networks, and we use the statistical viewpoint to highlight strengths and weaknesses of neural models. We illustrate the main points with some recognition experiments involving artificial data as well as handwritten numerals. In way of conclusion, we suggest that current-generation feedforward neural networks are largely inadequate for difficult problems in machine perception and machine learning, regardless of parallel-versus-serial hardware or other implementation issues. Furthermore, we suggest that the fundamental challenges in neural modeling are about representation rather than learning per se. This last point is supported by additional experiments with handwritten numerals.

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3. A Model for the Action of NMDA Conductances in the Visual Cortex

Kevin Fox, Nigel W. Daw

https://doi.org/10.1162/neco.1992.4.1.59 | pp. 59-83

A model is presented for the behavior of N-methyl-D-aspartate (NMDA) receptors during neuronal responses to visual stimuli in visual cortex based on classical steady-state equations for membrane conductance, and for binding of transmitter with NMDA and non-NMDA receptors. Constraints in the equations and the equation for voltage dependency of the NMDA receptors come from measurements in hippocampus, embryonic CNS neurons, and cortex. An excitatory amino acid transmitter released from terminals in the visual cortex is assumed to be related to the contrast of the stimulus by a hyperbolic tangent formula. The transmitter is assumed to be glutamate. The model is a two compartment representation of a single neuron at steady state. The model fits results obtained from contrast-response curves measured in the visual cortex in control conditions and with iontophoresis of NMDA and non-NMDA agonists or antagonists. The model shows how NMDA receptors contribute to the visual response in a graded multiplicative fashion at all levels of contrast. NMDA receptors do not show switch behavior, that is, they are not turned on at high levels of stimulation only. The difference in NMDA receptor current between low and high levels of stimulation cannot be accounted for solely by the voltage dependency of the NMDA receptor. One needs, in addition, another factor, such as a difference in the Hill coefficient for binding of glutamate at

the NMDA receptor.

4. Alternating and Synchronous Rhythms in Reciprocally Inhibitory Model Neurons

Xiao■Jing Wang, John Rinzel

https://doi.org/10.1162/neco.1992.4.1.84 | pp. 84-97

We study pacemaker rhythms generated by two nonoscillatory model cells that are coupled by inhibitory synapses. A minimal ionic model that exhibits postinhibitory rebound (PIR) is presented. When the post-synaptic conductance depends instantaneously on presynaptic potential the classical alternating rhythm is obtained. Using phase-plane analysis we identify two underlying mechanisms, “release” and “escape,” for the out-of-phase oscillation. When the postsynaptic conductance is not instantaneous but decays slowly, the two cells can oscillate synchronously with no phase difference. In each case, different stable activity patterns can coexist over a substantial parameter range.

5. Feature Extraction Using an Unsupervised Neural Network

Nathan Intrator

https://doi.org/10.1162/neco.1992.4.1.98 | pp. 98-107

A novel unsupervised neural network for dimensionality reduction that seeks directions emphasizing multimodality is presented, and its connection to exploratory projection pursuit methods is discussed. This leads to a new statistical insight into the synaptic modification equations governing learning in Bienenstock, Cooper, and Munro (BCM) neurons (1982). The importance of a dimensionality reduction principle based solely on distinguishing features is demonstrated using a phoneme recognition experiment. The extracted features are compared with features extracted using a backpropagation network.

6. Speaker-Independent Digit Recognition Using a Neural Network with Time-Delayed Connections

K. P. Unnikrishnan, J. J. Hopfield, David W. Tank

https://doi.org/10.1162/neco.1992.4.1.108 | pp. 108-119

The capability of a small neural network to perform speaker-independent recognition of spoken digits in connected speech has been investigated. The network uses time delays to organize rapidly changing outputs of symbol detectors over the time scale of a word. The network is data driven and unclocked. To achieve useful accuracy in a speaker-independent setting, many new ideas and procedures were developed. These include improving the feature detectors, self-recognition of word ends, reduction in network size, and dividing speakers into natural classes. Quantitative experiments based on Texas Instruments (TI) digit databases are described.

7. Local Feedback Multilayered Networks

Paolo Frasconi, Marco Gori, Giovanni Soda
https://doi.org/10.1162/neco.1992.4.1.120 | pp. 120-130

In this paper, we investigate the capabilities of local feedback multilayered networks, a particular class of recurrent networks, in which feedback connections are only allowed from neurons to themselves. In this class, learning can be accomplished by an algorithm that is local in both space and time. We describe the limits and properties of these networks and give some insights on their use for solving practical problems.

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8. Learning to Control Fast-Weight Memories: An Alternative to Dynamic Recurrent Networks

Jürgen Schmidhuber
https://doi.org/10.1162/neco.1992.4.1.131 | pp. 131-139

Previous algorithms for supervised sequence learning are based on dynamic recurrent networks. This paper describes an alternative class of gradient-based systems consisting of two feedforward nets that learn to deal with temporal sequences using fast weights: The first net learns to produce context-dependent weight changes for the second net whose weights may vary very quickly. The method offers the potential for STM storage efficiency: A single weight (instead of a full-fledged unit) may be sufficient for storing temporal information. Various learning methods are derived. Two experiments with unknown time delays illustrate the approach. One experiment shows how the system can be used for adaptive temporary variable binding.

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9. First- and Second-Order Methods for Learning: Between Steepest Descent and Newton's Method

Roberto Battiti
https://doi.org/10.1162/neco.1992.4.2.141 | pp. 141-166

On-line first-order backpropagation is sufficiently fast and effective for many large-scale classification problems but for very high precision mappings, batch processing may be the method of choice. This paper reviews first- and second-order optimization methods for learning in feedforward neural networks. The viewpoint is that of optimization: many methods can be cast in the language of optimization techniques, allowing the transfer to neural nets of detailed results about computational complexity and safety procedures to ensure convergence and to avoid numerical problems. The review is not intended to deliver detailed prescriptions for the most appropriate methods in specific applications, but to illustrate the main characteristics of the different methods and their mutual relations.

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10. Efficient Simplex-Like Methods for Equilibria of Nonsymmetric Analog Networks

Douglas A. Miller, Steven W. Zucker
https://doi.org/10.1162/neco.1992.4.2.167 | pp. 167-190

What is the complexity of computing equilibria for physically implementable analog networks (Hopfield 1984; Sejnowski 1981) with arbitrary connectivity? We show that if the amplifiers are piecewise-linear, then such

networks are instances of a game-theoretic model known as polymatrix games. In contrast with the usual gradient descent methods for symmetric networks, equilibria for polymatrix games may be computed by vertex pivoting algorithms similar to the simplex method for linear programming. Like the simplex method, these algorithms have characteristic low order polynomial behavior in virtually all practical cases, though not certain theoretical ones. While these algorithms cannot be applied to models requiring evolution from an initial point, they are applicable to “clamping” models whose input is expressed purely as a bias. Thus we have an a priori indication that such models are computationally tractable.

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11. A Volatility Measure for Annealing in Feedback Neural Networks

J. Alspector, T. Zeppenfeld, Stephan Luna

https://doi.org/10.1162/neco.1992.4.2.191 | pp. 191-195

In feedback neural networks, especially for static pattern learning, a reliable method of settling is required. Simulated annealing has been used but it is often difficult to determine how to set the annealing schedule. Often the specific heat is used as a measure of when to slow down the annealing process, but this is difficult to measure. We propose another measure, volatility, which is easy to measure and related to the Edwards-Anderson model in spin-glass physics. This paper presents the concept of volatility, an argument for its similarity to specific heat, simulations of dynamics in Boltzmann and mean-field networks, and a method of using it to speed up learning.

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12. What Does the Retina Know about Natural Scenes?

Joseph J. Atick, Amanda Redlich

https://doi.org/10.1162/neco.1992.4.2.196 | pp. 196-210

By examining the experimental data on the statistical properties of natural scenes together with (retinal) contrast sensitivity data, we arrive at a first principle, theoretical hypothesis for the purpose of retinal processing and its relationship to an animal's environment. We argue that the retinal goal is to transform the visual input as much as possible into a statistically independent basis as the first step in creating a redundancy reduced representation in the cortex, as suggested by Barlow. The extent of this whitening of the input is limited, however, by the need to suppress input noise. Our explicit theoretical solutions for the retinal filters also show a simple dependence on mean stimulus luminance: they predict an approximate Weber law at low spatial frequencies and a De Vries-Rose law at high frequencies. Assuming that the dominant source of noise is quantum, we generate a family of contrast sensitivity curves as a function of mean luminance. This family is compared to psychophysical data.

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13. A Simple Network Showing Burst Synchronization without Frequency Locking

Christof Koch, Heinz Georg Schuster

https://doi.org/10.1162/neco.1992.4.2.211 | pp. 211-223

The dynamic behavior of a network model consisting of all-to-all excitatory coupled binary neurons with global inhibition is studied analytically and numerically. We prove that for random input signals, the output of

the network consists of synchronized bursts with apparently random intermissions of noisy activity. We introduce the fraction of simultaneously firing neurons as a measure for synchrony and prove that its temporal correlation function displays, besides a delta peak at zero indicating random processes, strongly dampened oscillations. Our results suggest that synchronous bursts can be generated by a simple neuronal architecture that amplifies incoming coincident signals. This synchronization process is accompanied by dampened oscillations that, by themselves, however, do not play any constructive role in this and can therefore be considered to be an epiphenomenon.

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14. On a Magnitude Preserving Iterative MAXnet Algorithm

Bruce W. Suter, Matthew Kabrisky

https://doi.org/10.1162/neco.1992.4.2.224 | pp. 224-233

A new iterative maximum picking neural net (MAXnet) is presented. This formulation determines the value and the location either for a unique maximum or for multiple maxima. This new net converges, for many commonly occurring distributions, in O(log M) iterations using only simple computing elements.

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15. Learning Complex, Extended Sequences Using the Principle of History Compression

Jürgen Schmidhuber

https://doi.org/10.1162/neco.1992.4.2.234 | pp. 234-242

Previous neural network learning algorithms for sequence processing are computationally expensive and perform poorly when it comes to long time lags. This paper first introduces a simple principle for reducing the descriptions of event sequences without loss of information. A consequence of this principle is that only unexpected inputs can be relevant. This insight leads to the construction of neural architectures that learn to “divide and conquer” by recursively decomposing sequences. I describe two architectures. The first functions as a self-organizing multilevel hierarchy of recurrent networks. The second, involving only two recurrent networks, tries to collapse a multilevel predictor hierarchy into a single recurrent net. Experiments show that the system can require less computation per time step and many fewer training sequences than conventional training algorithms for recurrent nets.

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16. A Fixed Size Storage O(n³) Time Complexity Learning Algorithm for Fully Recurrent Continually Running Networks

Jürgen Schmidhuber

https://doi.org/10.1162/neco.1992.4.2.243 | pp. 243-248

The real-time recurrent learning (RTRL) algorithm (Robinson and Fallside 1987; Williams and Zipser 1989) requires O(n 4) computations per time step, where n is the number of noninput units. I describe a method suited for on-line learning that computes exactly the same gradient and requires fixed-size storage of the same order but has an average time complexity per time step of O(n 3).

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17. How Tight Are the Vapnik-Chervonenkis Bounds?

David Cohn, Gerald Tesauro

https://doi.org/10.1162/neco.1992.4.2.249 | pp. 249-269

We describe a series of numerical experiments that measure the average generalization capability of neural networks trained on a variety of simple functions. These experiments are designed to test the relationship between average generalization performance and the worst-case bounds obtained from formal learning theory using the Vapnik-Chervonenkis (VC) dimension (Blumer et al. 1989; Haussler et al. 1990). Recent statistical learning theories (Tishby et al. 1989; Schwartz et al. 1990) suggest that surpassing these bounds might be possible if the spectrum of possible generalizations has a “gap” near perfect performance. We indeed find that, in some cases, the average generalization is significantly better than the VC bound: the approach to perfect performance is exponential in the number of examples m, rather than the 1/m result of the bound. However, in these cases, we have not found evidence of the gap predicted by the above statistical theories. In other cases, we do find the 1/m behavior of the VC bound, and in these cases, the numerical prefactor is closely related to the prefactor contained in the bound.

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18. Working Memory Networks for Learning Temporal Order with Application to Three-Dimensional Visual Object Recognition

Gary Bradski, Gail A. Carpenter, Stephen Grossberg

https://doi.org/10.1162/neco.1992.4.2.270 | pp. 270-286

Working memory neural networks, called Sustained Temporal Order REcurrent (STORE) models, encode the invariant temporal order of sequential events in short-term memory (STM). Inputs to the networks may be presented with widely differing growth rates, amplitudes, durations, and interstimulus intervals without altering the stored STM representation. The STORE temporal order code is designed to enable groupings of the stored events to be stably learned and remembered in real time, even as new events perturb the system. Such invariance and stability properties are needed in neural architectures which self-organize learned codes for variable-rate speech perception, sensorimotor planning, or three-dimensional (3-D) visual object recognition. Using such a working memory, a self-organizing architecture for invariant 3-D visual object recognition is described. The new model is based on the model of Seibert and Waxman (1990a), which builds a 3-D representation of an object from a temporally ordered sequence of its two-dimensional (2-D) aspect graphs. The new model, called an ARTSTORE model, consists of the following cascade of processing modules: Invariant Preprocessor → ART 2 → STORE Model → ART 2 → Outstar Network.

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19. A Competitive Distribution Theory of Neocortical Dynamics

James A. Reggia, C. Lynne D’Autrechy, Granger Sutton, Michael Weinrich

https://doi.org/10.1162/neco.1992.4.3.287 | pp. 287-317

Peristimulus inhibition in sensory pathways is generally attributed to lateral inhibitory connections. However, in the neocortex circuitry is incompletely understood at present, and in some cases there is an apparent mismatch between observed inhibitory effects and intracortical inhibitory connections. This paper studies the hypothesis that an additional mechanism, competitive distribution of activation, underlies some inhibitory effects in cortex. Analysis of a mathematical model based on this hypothesis predicts that per stimulus

inhibitory effects can be caused by competitive distribution of activation, and computer simulations confirm these predictions by demonstrating Mexican Hat patterns of lateral interactions, transformation of initially diffuse activity patterns into tightly focused "islands" of activation, and edge enhancement. The amount of inhibition can be adjusted by varying the intensity of the underlying competitive process. The concept of competitive distribution of activation provides an important perspective for interpreting neocortical and thalamocortical circuitry and can serve as a guide for further morphological and physiological studies. For example, it provides an explanation for the existence of recurrent cortex-to-thalamus connections that perform a logical AND-operation, and predicts the existence of analogous neocortical circuitry.

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20. A Network Simulation of Thalamic Circuit Operations in Selective Attention

David LaBerge, Marc Carter, Vincent Brown

https://doi.org/10.1162/neco.1992.4.3.318 | pp. 318-331

The ability of a thalamic circuit to process information selectively from a spatial location was investigated in neural network models. Starting with the known general structure of the thalamic circuit, we considered three variations of the projections from the inhibitory cells of the reticular nucleus onto the cells of the pulvinar nucleus of the dorsal thalamus. The three circuits were modeled as systems of difference equations, and their operations were simulated by computer-based numerical integration. In all three circuits, when input from a target location was slightly larger than the input from neighboring locations, the time evolution of principal (relay) cell outputs showed substantial selective enhancement at the target location compared with neighboring locations. The selective enhancement effect was produced not only on ascending inputs but also on descending cortical inputs. Simulations separating the lateral inhibitory and feedback-enhancement components of the circuits suggested that the feedback-enhancement component substantially magnified the ability of lateral inhibition to produce a target/surround difference.

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21. Generation of Direction Selectivity by Isotropic Intracortical Connections

Florentin Wörgötter, Ernst Niebur, Christof Koch

https://doi.org/10.1162/neco.1992.4.3.332 | pp. 332-340

To what extent do the mechanisms generating different receptive field properties of neurons depend on each other? We investigated this question theoretically within the context of orientation and direction tuning of simple cells in the mammalian visual cortex. In our model a cortical cell of the "simple" type receives its orientation tuning by afferent convergence of aligned receptive fields of the lateral geniculate nucleus (Hubel and Wiesel 1962). We sharpen this orientation bias by postulating a special type of radially symmetric long-range lateral inhibition called circular inhibition. Surprisingly, this isotropic mechanism leads to the emergence of a strong bias for the direction of motion of a bar. We show that this directional anisotropy is neither caused by the probabilistic nature of the connections nor is it a consequence of the specific columnar structure chosen but that it is an inherent feature of the architecture of visual cortex.

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22. Binding Hierarchies: A Basis for Dynamic Perceptual Grouping

Erik D. Lumer, Bernardo A. Huberman
https://doi.org/10.1162/neco.1992.4.3.341 | pp. 341-355

Since it has been suggested that the brain binds its fragmentary representations of perceptual events via phase-locking of stimulated neural oscillators, it is important to determine how extended synchronization can occur in a clustered organization of cells possessing a distribution of firing rates. To answer that question, we establish the basic conditions for the existence of a binding mechanism based on synchronized oscillations. In addition, we present a simple hierarchical architecture of feedback units that not only induces robust phase-locking within and segregation between perceptual groups, but also serves as a generic binding machine.

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23. A Distributed Neural Network Architecture for Hexapod Robot Locomotion

Randall D. Beer, Hillel J. Chiel, Roger D. Quinn, Kenneth S. Espenschied, Patrik Larsson
https://doi.org/10.1162/neco.1992.4.3.356 | pp. 356-365

We present fully distributed neural network architecture for controlling the locomotion of a hexapod robot. The design of this network is directly based on work on the neuroethology of insect locomotion. Previously, we demonstrated in simulation that this controller could generate a continuous range of statically stable insect-like gaits as the activity of a single command neuron was varied and that it was robust to a variety of lesions. We now report that the controller can be utilized to direct the locomotion of an actual six-legged robot, and that it exhibits a range of gaits and degree of robustness in the real world that is quite similar to that observed in simulation.

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24. Multilayer Perceptron Learning Optimized for On-Chip Implementation: A Noise-Robust System

Alan F. Murray
https://doi.org/10.1162/neco.1992.4.3.366 | pp. 366-381

1 Requirement This paper describes an approach to multilayer perceptron (MLP) learning that is optimized for hardware implementation. Experimental results to date are promising, and it is the express aim of this paper to present these results concisely - a detailed mathematical analysis will follow in a subsequent, longer publication. Error backpropagation (Rumelhart et al. 1986) has achieved remarkable success as an algorithm for solving hard classification problems with MLP networks. It is, however, not readily amenable to VLSI integration, and the distinction it draws between hidden and output nodes renders it hostile to analog circuit forms. Use of the mathematical chain rule, to calculate the effect of a weight connecting to a hidden unit on the errors {Q} in the output layer, renders the error calculation scheme for hidden units different from, and more complicated than that for output units. The Virtual Targets learning scheme circumvents this problem by introducing an explicit desired state, or target for each of the hidden units, which is updated continuously, and stored along with the synapse weights. While this means that a target state must be stored for each input pattern and hidden node, it simplifies and renders homogeneous the process of weight evolution for all neurons. Furthermore, since a target state is already stored for each output neuron, the scheme essentially removes the distinction during learning between hidden and output nodes. Analog

integrated circuits based on the virtual targets strategy will therefore be flexible in architectural terms, as all units will be configurable as either output or hidden layer neurons. The fundamental idea of adapting the internal representation, either as well as, or instead of the weights, is not itself new (Rohwer 1990; Grossman et al. 1990; Krogh et al. 1990). However, these pieces of work were not optimized for hardware implementation. The fundamental difference is that simplicity of implementation has been made the primary goal in the work described in this paper, to produce a system optimized for analog VLSI. There are also several important differences in detail between the work described in this paper and these earlier, similar Neural Computation 4, 366-381 (1992) @ 1992 Massachusetts Institute of Technology

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25. Computing the Karhunen-Loeve Expansion with a Parallel, Unsupervised Filter System

Reiner Lenz, Mats Österberg

https://doi.org/10.1162/neco.1992.4.3.382 | pp. 382-392

We use the invariance principle and the principles of maximum information extraction and maximum signal concentration to design a parallel, linear filter system that learns the Karhunen-Loeve expansion of a process from examples. In this paper we prove that the learning rule based on these principles forces the system into stable states that are pure eigenfunctions of the input process.

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26. Learning and Extracting Finite State Automata with Second-Order Recurrent Neural Networks

C. Lee Giles, CLIFFORD B. MILLER, Dong Chen, H. H. Chen, Guo-Zheng Sun, Y. C. Lee

https://doi.org/10.1162/neco.1992.4.3.393 | pp. 393-405

We show that a recurrent, second-order neural network using a real-time, forward training algorithm readily learns to infer small regular grammars from positive and negative string training samples. We present simulations that show the effect of initial conditions, training set size and order, and neural network architecture. All simulations were performed with random initial weight strengths and usually converge after approximately a hundred epochs of training. We discuss a quantization algorithm for dynamically extracting finite state automata during and after training. For a well-trained neural net, the extracted automata constitute an equivalence class of state machines that are reducible to the minimal machine of the inferred grammar. We then show through simulations that many of the neural net state machines are dynamically stable, that is, they correctly classify many long unseen strings. In addition, some of these extracted automata actually outperform the trained neural network for classification of unseen strings.

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27. Induction of Finite-State Languages Using Second-Order Recurrent Networks

Raymond L. Watrous, Gary M. Kuhn

https://doi.org/10.1162/neco.1992.4.3.406 | pp. 406-414

Second-order recurrent networks that recognize simple finite state languages over {0,1}* are induced from positive and negative examples. Using the complete gradient of the recurrent network and sufficient training

examples to constrain the definition of the language to be induced, solutions are obtained that correctly recognize strings of arbitrary length.

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28. Bayesian Interpolation

David Mackay

https://doi.org/10.1162/neco.1992.4.3.415 | pp. 415-447

Although Bayesian analysis has been in use since Laplace, the Bayesian method of model-comparison has only recently been developed in depth. In this paper, the Bayesian approach to regularization and model-comparison is demonstrated by studying the inference problem of interpolating noisy data. The concepts and methods described are quite general and can be applied to many other data modeling problems. Regularizing constants are set by examining their posterior probability distribution. Alternative regularizers (priors) and alternative basis sets are objectively compared by evaluating the evidence for them. “Occam's razor” is automatically embodied by this process. The way in which Bayes infers the values of regularizing constants and noise levels has an elegant interpretation in terms of the effective number of parameters determined by the data set. This framework is due to Gull and Skilling.

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29. A Practical Bayesian Framework for Backpropagation Networks

David Mackay

https://doi.org/10.1162/neco.1992.4.3.448 | pp. 448-472

A quantitative and practical Bayesian framework is described for learning of mappings in feedforward networks. The framework makes possible (1) objective comparisons between solutions using alternative network architectures, (2) objective stopping rules for network pruning or growing procedures, (3) objective choice of magnitude and type of weight decay terms or additive regularizers (for penalizing large weights, etc.), (4) a measure of the effective number of well-determined parameters in a model, (5) quantified estimates of the error bars on network parameters and on network output, and (6) objective comparisons with alternative learning and interpolation models such as splines and radial basis functions. The Bayesian "evidence" automatically embodies "Occam's razor," penalizing overflexible and overcomplex models. The Bayesian approach helps detect poor underlying assumptions in learning models. For learning models well matched to a problem, a good correlation between generalization ability and the Bayesian evidence is obtained.

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30. Simplifying Neural Networks by Soft Weight-Sharing

Steven J. Nowlan, Geoffrey E. Hinton

https://doi.org/10.1162/neco.1992.4.4.473 | pp. 473-493

One way of simplifying neural networks so they generalize better is to add an extra term to the error function that will penalize complexity. Simple versions of this approach include penalizing the sum of the squares of the weights or penalizing the number of nonzero weights. We propose a more complicated penalty term in which the distribution of weight values is modeled as a mixture of multiple gaussians. A set of weights is simple if the weights have high probability density under the mixture model. This can be achieved by

clustering the weights into subsets with the weights in each cluster having very similar values. Since we do not know the appropriate means or variances of the clusters in advance, we allow the parameters of the mixture model to adapt at the same time as the network learns. Simulations on two different problems demonstrate that this complexity term is more effective than previous complexity terms.

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31. Exact Calculation of the Hessian Matrix for the Multilayer Perceptron

Chris Bishop

https://doi.org/10.1162/neco.1992.4.4.494 | pp. 494-501

The elements of the Hessian matrix consist of the second derivatives of the error measure with respect to the weights and thresholds in the network. They are needed in Bayesian estimation of network regularization parameters, for estimation of error bars on the network outputs, for network pruning algorithms, and for fast retraining of the network following a small change in the training data. In this paper we present an extended backpropagation algorithm that allows all elements of the Hessian matrix to be evaluated exactly for a feedforward network of arbitrary topology. Software implementation of the algorithm is straightforward.

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32. NMDA-Based Pattern Discrimination in a Modeled Cortical Neuron

Bartlett W. Mel

https://doi.org/10.1162/neco.1992.4.4.502 | pp. 502-517

Compartmental simulations of an anatomically characterized cortical pyramidal cell were carried out to study the integrative behavior of a complex dendritic tree. Previous theoretical (Feldman and Ballard 1982; Durbin and Rumelhart 1989; Mel 1990; Mel and Koch 1990; Poggio and Girosi 1990) and compartmental modeling (Koch et al. 1983; Shepherd et al. 1985; Koch and Poggio 1987; Rall and Segev 1987; Shepherd and Brayton 1987; Shepherd et al. 1989; Brown et al. 1991) work had suggested that multiplicative interactions among groups of neighboring synapses could greatly enhance the processing power of a neuron relative to a unit with only a single global firing threshold. This issue was investigated here, with a particular focus on the role of voltage-dependent N-methyl-D-asparate (NMDA) channels in the generation of cell responses. First, it was found that when a large proportion of the excitatory synaptic input to dendritic spines is carried by NMDA channels, the pyramidal cell responds preferentially to spatially clustered, rather than random, distributions of activated synapses. Second, based on this mechanism, the NMDA-rich neuron is shown to be capable of solving a nonlinear pattern discrimination task. We propose that manipulation of the spatial ordering of afferent synaptic connections onto the dendritic arbor is a possible biological strategy for pattern information storage during learning.

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33. The Impact of Parallel Fiber Background Activity on the Cable Properties of Cerebellar Purkinje Cells

Moshe Rapp, Yosef Yarom, Idan Segev

https://doi.org/10.1162/neco.1992.4.4.518 | pp. 518-533

Neurons in the mammalian CNS receive 10 4 -10 5 synaptic inputs onto their dendritic tree. Each of these inputs may fire spontaneously at a rate of a few spikes per second. Consequently, the cell is bombarded by

several hundred synapses in each and every millisecond. An extreme example is the cerebellar Purkinje cell (PC) receiving approximately 100,000 excitatory synapses from the parallel fibers (p.f.s) onto dendritic spines covering the thin dendritic branchlets. What is the effect of the p.f.s activity on the integrative capabilities of the PC? This question is explored theoretically using analytical cable models as well as compartmental models of a morphologically and physiologically characterized PC from the guinea pig cerebellum. The input of individual p.f.s was modeled as a transient conductance change, peaking at 0.4 nS with a rise time of 0.3 msec and a reversal potential of +60 mV relative to rest. We found that already at a firing frequency of a few spikes per second the membrane conductance is several times larger than the membrane conductance in the absence of synaptic activity. As a result, the cable properties of the PC significantly change; the most sensitive parameters are the system time constant (τ 0) and the steady-state attenuation factor from dendritic terminal to soma. The implication is that the cable properties of central neurons in freely behaving animals are different from those measured in slice preparation or in anesthetized animals, where most of the synaptic inputs are inactive. We conclude that, because of the large conductance increase produced by the background activity of the p.f.s, the activity of the PC will be altered from this background level either when the p.f.s change their firing frequency for a period of several tens of milliseconds or when a large population of the p.f.s fires during a narrow time window.

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34. Activity Patterns of a Slow Synapse Network Predicted by Explicitly Averaging Spike Dynamics

John Rinzel, Paul Frankel

https://doi.org/10.1162/neco.1992.4.4.534 | pp. 534-545

When postsynaptic conductance varies slowly compared to the spike generation process, a straightforward averaging scheme can be used to reduce the system's complexity. Our model consists of a Hodgkin-Huxley-like membrane description for each cell; synaptic activation is described by first order kinetics, with slow rates, in which the equilibrium activation is a sigmoidal function of the presynaptic voltage. Our work concentrates on a two-cell network and it applies qualitatively to the activity patterns, including bistable behavior, recently observed in simple in vitro circuits with slow synapses (Kleinfeld et al. 1990). The fact that our averaged system is derived from a realistic biophysical model has important consequences. In particular, it can preserve certain hysteresis behavior near threshold that is not represented in a simple ad hoc sigmoidal input-output network. This behavior enables a coupled pair of cells, one excitatory and one inhibitory, to generate an alternating burst rhythm even though neither cell has fatiguing properties.

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35. Phase Coupling in Simulated Chains of Coupled Oscillators Representing the Lamprey Spinal Cord

Thelma L. Williams

https://doi.org/10.1162/neco.1992.4.4.546 | pp. 546-558

Previous application of a mathematical theory of chains of coupled oscillators to the results of experiments on the lamprey spinal cord led to conclusions about the mechanisms of intersegmental coordination in the lamprey. The theory provides no direct link, however, to electrophysiological data obtained at the cellular level, nor are the details of the neuronal circuitry in the lamprey known. In this paper, a variant of the theory is developed for which the relevant variables can potentially be measured. This theory will be applied to

measurements on simulated oscillators, based on a network that has been postulated to constitute the basic circuitry of the segmental oscillator in the lamprey. A linear approximation to the equations is derived, and it will be shown that the behavior of simulated chains of these oscillators obeys the predictions of this approximation.

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36. Understanding Retinal Color Coding from First Principles

Joseph J. Atick, Li Zhaoping, Amanda Redlich

https://doi.org/10.1162/neco.1992.4.4.559 | pp. 559-572

A previously proposed theory of visual processing, based on redundancy reduction, is used to derive the retinal transfer function including color. The predicted kernels show the nontrivial mixing of space-time with color coding observed in experiments. The differences in color-coding between species are found to be due to differences among the chromatic autocorrelators for natural scenes in different environments.

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37. Interaction between Transparency and Structure from Motion

Daniel Kersten, HH Bülthoff, Bennett L. Schwartz, Kenneth J. Kurtz

https://doi.org/10.1162/neco.1992.4.4.573 | pp. 573-589

It is well known that the human visual system can reconstruct depth from simple random-dot displays given binocular disparity or motion information. This fact has lent support to the notion that stereo and structure from motion systems rely on low-level primitives derived from image intensities. In contrast, the judgment of surface transparency is often considered to be a higher-level visual process that, in addition to pictorial cues, utilizes stereo and motion information to separate the transparent from the opaque parts. We describe a new illusion and present psychophysical results that question this sequential view by showing that depth from transparency and opacity can override the bias to see rigid motion. The brain's computation of transparency may involve a two-way interaction with the computation of structure from motion.

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38. Information-Based Objective Functions for Active Data Selection

David Mackay

https://doi.org/10.1162/neco.1992.4.4.590 | pp. 590-604

Learning can be made more efficient if we can actively select particularly salient data points. Within a Bayesian learning framework, objective functions are discussed that measure the expected informativeness of candidate measurements. Three alternative specifications of what we want to gain information about lead to three different criteria for data selection. All these criteria depend on the assumption that the hypothesis space is correct, which may prove to be their main weakness.

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39. Four Types of Learning Curves

■■■-■■■ ■■■■■, *Naotake FUJITA, Shigeru Shinomoto*

https://doi.org/10.1162/neco.1992.4.4.605 | pp. 605-618

If machines are learning to make decisions given a number of examples, the generalization error ε(t) is defined as the average probability that an incorrect decision is made for a new example by a machine when trained with t examples. The generalization error decreases as t increases, and the curve ε(t) is called a learning curve. The present paper uses the Bayesian approach to show that given the annealed approximation, learning curves can be classified into four asymptotic types. If the machine is deterministic with noiseless teacher signals, then (1) ε ~ at -1 when the correct machine parameter is unique, and (2) ε ~ at -2 when the set of the correct parameters has a finite measure. If the teacher signals are noisy, then (3) ε ~ at -1/2 for a deterministic machine, and (4) ε ~ c + at -1 for a stochastic machine.

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40. Errata

(authors unknown)

https://doi.org/10.1162/neco.1992.4.4.619 | pp. 619-619

No abstract available.

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41. Nonlinear Dynamics and Symbolic Dynamics of Neural Networks

John E. Lewis, Leon Glass

https://doi.org/10.1162/neco.1992.4.5.621 | pp. 621-642

A piecewise linear equation is proposed as a method of analysis of mathematical models of neural networks. A symbolic representation of the dynamics in this equation is given as a directed graph on an N-dimensional hypercube. This provides a formal link with discrete neural networks such as the original Hopfield models. Analytic criteria are given to establish steady states and limit cycle oscillations independent of network dimension. Model networks that display multiple stable limit cycles and chaotic dynamics are discussed. The results show that such equations are a useful and efficient method of investigating the behavior of neural networks.

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42. Cortical Cells Should Fire Regularly, But Do Not

William R. Softky, Christof Koch

https://doi.org/10.1162/neco.1992.4.5.643 | pp. 643-646

When a typical nerve cell is injected with enough current, it fires a regular stream of action potentials. But cortical cells in vivo usually fire irregularly, reflecting synaptic input from presynaptic cells as well as intrinsic biophysical properties. We have applied the theory of stochastic
\nprocesses to spike trains recorded from cortical neurons (Tuckwell 1989) and find a fundamental contradiction between the large interspike variability observed and the much lower values predicted by well-accepted biophysical models of single cells.

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43. A Simplified Neural-Network Solution through Problem Decomposition: The Case of the Truck Backer-Upper

Robert E. Jenkins, Ben Yuhas

https://doi.org/10.1162/neco.1992.4.5.647 | pp. 647-649

September 01 1992 A Simplified Neural-Network Solution through Problem Decomposition: The Case of the Truck Backer-Upper In Special Collection: CogNet Robert E. Jenkins, Robert E. Jenkins The Applied Physics Laboratory, The Johns Hopkins University, Baltimore, MD 21218 USA Search for other works by this author on: This Site Google Scholar Ben P. Yuhas Ben P. Yuhas The Applied Physics Laboratory, The Johns Hopkins University, Baltimore, MD 21218 USA Search for other works by this author on: This Site Google Scholar Author and Article Information Robert E. Jenkins The Applied Physics Laboratory, The Johns Hopkins University, Baltimore, MD 21218 USA Ben P. Yuhas The Applied Physics Laboratory, The Johns Hopkins University, Baltimore, MD 21218 USA *Current address: Bellcore MRE 2E-330, Morristown, NJ 07962-1910 USA. Received: December 17 1991 Accepted: March 04 1992 Online ISSN: 1530-888X Print ISSN: 0899-7667 © 1992 Massachusetts Institute of Technology1992 Neural Computation (1992) 4 (5): 647–649. https://doi.org/10.1162/neco.1992.4.5.647 Article history Received: December 17 1991 Accepted: March 04 1992 Cite Icon Cite Permissions Share Icon Share Facebook Twitter LinkedIn MailTo Views Icon Views Article contents Figures & tables Video Audio Supplementary Data Peer Review Search Site Citation Robert E. Jenkins, Ben P. Yuhas; A Simplified Neural-Network Solution through Problem Decomposition: The Case of the Truck Backer-Upper. Neural Comput 1992; 4 (5): 647–649. doi: https://doi.org/10.1162/neco.1992.4.5.647 Download citation file: Ris (Zotero) Reference Manager EasyBib Bookends Mendeley Papers EndNote RefWorks BibTex toolbar search Search Dropdown Menu toolbar search search input Search input auto suggest filter your search All ContentAll JournalsNeural Computation Search Advanced Search This content is only available as a PDF. © 1992 Massachusetts Institute of Technology1992 Article PDF first page preview Close Modal You do not currently have access to this content.

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44. Learning to Segment Images Using Dynamic Feature Binding

Michael C. Mozer, Richard S. Zemel, Marlene Behrmann, Christopher K. I. Williams

https://doi.org/10.1162/neco.1992.4.5.650 | pp. 650-665

Despite the fact that complex visual scenes contain multiple, overlapping objects, people perform object recognition with ease and accuracy. One operation that facilitates recognition is an early segmentation process in which features of objects are grouped and labeled according to which object they belong. Current computational systems that perform this operation are based on predefined grouping heuristics. We describe a system called MAGIC that learns how to group features based on a set of presegmented examples. In many cases, MAGIC discovers grouping heuristics similar to those previously proposed, but it also has the capability of finding nonintuitive structural regularities in images. Grouping is performed by a relaxation network that attempts to dynamically bind related features. Features transmit a complex-valued signal (amplitude and phase) to one another; binding can thus be represented by phase locking related features. MAGIC's training procedure is a generalization of recurrent backpropagation to complex-valued units.

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45. Stimulus-Dependent Assembly Formation of Oscillatory Responses: III. Learning

Peter König, Bernd Janosch, Thomas B. Schillen
https://doi.org/10.1162/neco.1992.4.5.666 | pp. 666-681

A temporal structure of neuronal activity has been suggested as a potential mechanism for defining cell assemblies in the brain. This concept has recently gained support by the observation of stimulus-dependent oscillatory activity in the visual cortex of the cat. Furthermore, experimental evidence has been found showing the formation and segregation of synchronously oscillating cell assemblies in response to various stimulus conditions. In previous work, we have demonstrated that a network of neuronal oscillators coupled by synchronizing and desynchronizing delay connections can exhibit a temporal structure of responses, which closely resembles experimental observations. In this paper, we investigate the self-organization of synchronizing and desynchronizing coupling connections by local learning rules. Based on recent experimental observations, we modify synchronizing connections according to a two-threshold learning rule, involving synaptic potentiation and depression. This rule is generalized to its functional inverse for weight changes of desynchronizing connections. We show that after training, the resulting network exhibits stimulus-dependent formation and segregation of oscillatory assemblies in agreement with the experimental data. These results indicate that local learning rules during ontogenesis can suffice to develop a connectivity pattern in support of the observed temporal structure of stimulus responses in cat visual cortex.

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46. Seeing Beyond the Nyquist Limit

Daniel Ruderman, William Bialek
https://doi.org/10.1162/neco.1992.4.5.682 | pp. 682-690

In many biological systems the primary transduction of sensory stimuli occurs in a regular array of receptors. Because of this discrete sampling it is usually assumed that the organism has no knowledge of signals beyond the Nyquist frequency. In fact, higher frequency signals are expected to mask the available lower frequency information as a result of aliasing. It has been suggested that these considerations are important in understanding, for example, the design of the receptor lattice in the mammalian fovea. We show that if the organism has knowledge of the probability distribution from which the signals are drawn, outputs from a discrete receptor array can be used to estimate signals beyond the Nyquist limit. In effect, a priori knowledge can be used to de-alias the image, and the estimated signal above the Nyquist cutoff is in fact coherent with the real signal at these high frequencies. We address initially the problem of stimulus reconstruction from a noisy receptor array responding to a Gaussian stimulus ensemble. In this case, the best reconstruction strategy is a simple linear transformation. In the more interesting (and natural) case of nongaussian stimuli, optimal reconstruction requires nonlinear operations, but the higher order correlations in the stimulus ensemble can be used to improve the estimate of super-Nyquist signals.

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47. Local Synaptic Learning Rules Suffice to Maximize Mutual Information in a Linear Network

Ralph Linsker
https://doi.org/10.1162/neco.1992.4.5.691 | pp. 691-702

A network that develops to maximize the mutual information between its output and the signal portion of its input (which is admixed with noise) is useful for extracting salient input features, and may provide a model

for aspects of biological neural network function. I describe a local synaptic Learning rule that performs stochastic gradient ascent in this information-theoretic quantity, for the case in which the input-output mapping is linear and the input signal and noise are multivariate gaussian. Feedforward connection strengths are modified by a Hebbian rule during a "learning" phase in which examples of input signal plus noise are presented to the network, and by an anti-Hebbian rule during an "unlearning" phase in which examples of noise alone are presented. Each recurrent lateral connection has two values of connection strength, one for each phase; these values are updated by an anti-Hebbian rule.

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48. On the Information Storage Capacity of Local Learning Rules

Günther Palm

https://doi.org/10.1162/neco.1992.4.5.703 | pp. 703-711

A simple relation between the storage capacity A for autoassociation and H for heteroassociation with a local learning rule is demonstrated: H = 2A. Both values are bounded by local learning bounds: A ≤ L A and H ≤ L H. L H = 2L A is evaluated numerically.

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49. Learning Curves for Error Minimum and Maximum Likelihood Algorithms

Yoshiyuki Kabashima, Shigeru Shinomoto

https://doi.org/10.1162/neco.1992.4.5.712 | pp. 712-719

For the problem of dividing the space originally partitioned by a blurred boundary, every learning algorithm can make the probability of incorrect prediction of an individual example ε decrease with the number of training examples t. We address here the question of how the asymptotic form of ε(t) as well as its limit of convergence reflect the choice of learning algorithms. The error minimum algorithm is found to exhibit rather slow convergence of ε(t) to its lower bound ε 0 , ε(t) - ε 0 ~ O(t -2/3). Even for the purpose of minimizing prediction error, the maximum likelihood algorithm can be utilized as an alternative. If the true probability distribution happens to be contained in the family of hypothetical functions, then the boundary estimated from the hypothetical distribution function eventually converges to the best choice. Convergence of the prediction error is then ε(t) - ε 0 ~ O(t -1). If the true distribution is not available from the algorithm, however, the boundary generally does not converge to the best choice, but instead ε(t) - ε 1 ~ ±O(t -1/2), where ε 1 > ε 0 > 0.

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50. The Evidence Framework Applied to Classification Networks

David Mackay

https://doi.org/10.1162/neco.1992.4.5.720 | pp. 720-736

Three Bayesian ideas are presented for supervised adaptive classifiers. First, it is argued that the output of a classifier should be obtained by marginalizing over the posterior distribution of the parameters; a simple approximation to this integral is proposed and demonstrated. This involves a "moderation" of the most probable classifier's outputs, and yields improved performance. Second, it is demonstrated that the Bayesian framework for model comparison described for regression models in MacKay (1992a,b) can also be applied to classification problems. This framework successfully chooses the magnitude of weight decay

terms, and ranks solutions found using different numbers of hidden units. Third, an information-based data selection criterion is derived and demonstrated within this framework.

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51. Rotor Neurons: Basic Formalism and Dynamics

Lars Gislén, Carsten Peterson, Bo Söderberg

https://doi.org/10.1162/neco.1992.4.5.737 | pp. 737-745

Rotor neurons are introduced to encode states living on the surface of a sphere in D dimensions. Such rotors can be regarded as continuous generalizations of binary (Ising) neurons. The corresponding mean field equations are derived, and phase transition properties based on linearized dynamics are given. The power of this approach is illustrated with an optimization problem—placing N identical charges on a sphere such that the overall repulsive energy is minimized. The rotor approach appears superior to other methods for this problem both with respect to solution quality and computational effort needed.

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52. Refining PID Controllers Using Neural Networks

Gary M. Scott, Jude Shavlik, W. Harmon Ray

https://doi.org/10.1162/neco.1992.4.5.746 | pp. 746-757

The KBANN (Knowledge-Based Artificial Neural Networks) approach uses neural networks to refine knowledge that can be written in the form of simple propositional rules. We extend this idea further by presenting the MANNCON (Multivariable Artificial Neural Network Control) algorithm by which the mathematical equations governing a PID (Proportional-Integral-Derivative) controller determine the topology and initial weights of a network, which is further trained using backpropagation. We apply this method to the task of controlling the outflow and temperature of a water tank, producing statistically significant gains in accuracy over both a standard neural network approach and a nonlearning PID controller. Furthermore, using the PID knowledge to initialize the weights of the network produces statistically less variation in test set accuracy when compared to networks initialized with small random numbers.

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53. Ultrasound Tomography Imaging of Defects Using Neural Networks

Denis Anthony, Evor L. Hines, D.A. Hutchins, J. Toby Mottram

https://doi.org/10.1162/neco.1992.4.5.758 | pp. 758-771

Simulations of ultrasound tomography demonstrated that artificial neural networks can solve the inverse problem in ultrasound tomography. A highly simplified model of ultrasound propagation was constructed, taking no account of refraction or diffraction, and using only longitudinal wave time of flight (TOF). TOF data were used as the network inputs, and the target outputs were the expected pixel maps, showing defects (gray scale coded) according to the velocity of the wave in the defect. The effects of varying resolution and defect velocity were explored. It was found that defects could be imaged using time of flight of ultrasonic rays.

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54. Improving the Accuracy of an Artificial Neural Network Using Multiple Differently Trained Networks

William G. Baxt

https://doi.org/10.1162/neco.1992.4.5.772 | pp. 772-780

When either detection rate (sensitivity) or false alarm rate (specificity) is optimized in an artificial neural network trained to identify myocardial infarction, the increase in the accuracy of one is always done at the expense of the accuracy of the other. To overcome this loss, two networks that were separately trained on populations of patients with different likelihoods of myocardial infarction were used in concert. One network was trained on clinical pattern sets derived from patients who had a low likelihood of myocardial infarction, while the other was trained on pattern sets derived from patients with a high likelihood of myocardial infarction. Unknown patterns were analyzed by both networks. If the output generated by the network trained on the low risk patients was below an empirically set threshold, this output was chosen as the diagnostic output. If the output was above that threshold, the output of the network trained on the high risk patients was used as the diagnostic output. The dual network correctly identified 39 of the 40 patients who had sustained a myocardial infarction and 301 of 306 patients who did not have a myocardial infarction for a detection rate (sensitivity) and false alarm rate (1-specificity) of 97.50 and 1.63%, respectively. A parallel control experiment using a single network but identical training information correctly identified 39 of 40 patients who had sustained a myocardial infarction and 287 of 306 patients who had not sustained a myocardial infarction ($p = 0.003$).

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55. Rule-Based Neural Networks for Classification and Probability Estimation

R.M. Goodman, Charles M. Higgins, John W. V. Miller, Padhraic Smyth

https://doi.org/10.1162/neco.1992.4.6.781 | pp. 781-804

In this paper we propose a network architecture that combines a rule-based approach with that of the neural network paradigm. Our primary motivation for this is to ensure that the knowledge embodied in the network is explicitly encoded in the form of understandable rules. This enables the network's decision to be understood, and provides an audit trail of how that decision was arrived at. We utilize an information theoretic approach to learning a model of the domain knowledge from examples. This model takes the form of a set of probabilistic conjunctive rules between discrete input evidence variables and output class variables. These rules are then mapped onto the weights and nodes of a feedforward neural network resulting in a directly specified architecture. The network acts as parallel Bayesian classifier, but more importantly, can also output posterior probability estimates of the class variables. Empirical tests on a number of data sets show that the rule-based classifier performs comparably with standard neural network classifiers, while possessing unique advantages in terms of knowledge representation and probability estimation.

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56. Complex Scheduling with Potts Neural Networks

Lars Gislén, Carsten Peterson, Bo Söderberg

https://doi.org/10.1162/neco.1992.4.6.805 | pp. 805-831

In a recent paper (Gislén et al. 1989) a convenient encoding and an efficient mean field algorithm for solving scheduling problems using a Potts neural network was developed and numerically explored on simplified and synthetic problems. In this work the approach is extended to realistic applications both with respect to problem complexity and size. This extension requires among other things the interaction of Potts neurons with different number of components. We analyze the corresponding linearized mean field equations with respect to estimating the phase transition temperature. Also a brief comparison with the linear programming approach is given. Testbeds consisting of generated problems within the Swedish high school system are solved efficiently with high quality solutions as results.

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57. Asymmetric Parallel Boltzmann Machines are Belief Networks

Radford M. Neal

https://doi.org/10.1162/neco.1992.4.6.832 | pp. 832-834

November 01 1992 Asymmetric Parallel Boltzmann Machines are Belief Networks In Special Collection: CogNet Radford M. Neal Radford M. Neal Department of Computer Science, University of Toronto, 10 King's College Road, Toronto, Canada M5S 1A4 Search for other works by this author on: This Site Google Scholar Author and Article Information Radford M. Neal Department of Computer Science, University of Toronto, 10 King's College Road, Toronto, Canada M5S 1A4 Received: March 23 1992 Accepted: April 14 1992 Online ISSN: 1530-888X Print ISSN: 0899-7667 © 1992 Massachusetts Institute of Technology1992 Neural Computation (1992) 4 (6): 832–834. https://doi.org/10.1162/neco.1992.4.6.832 Article history Received: March 23 1992 Accepted: April 14 1992 Cite Icon Cite Permissions Share Icon Share Facebook Twitter LinkedIn MailTo Views Icon Views Article contents Figures & tables Video Audio Supplementary Data Peer Review Search Site Citation Radford M. Neal; Asymmetric Parallel Boltzmann Machines are Belief Networks. Neural Comput 1992; 4 (6): 832–834. doi: https://doi.org/10.1162/neco.1992.4.6.832 Download citation file: Ris (Zotero) Reference Manager EasyBib Bookends Mendeley Papers EndNote RefWorks BibTex toolbar search Search Dropdown Menu toolbar search search input Search input auto suggest filter your search All ContentAll JournalsNeural Computation Search Advanced Search This content is only available as a PDF. © 1992 Massachusetts Institute of Technology1992 Article PDF first page preview Close Modal You do not currently have access to this content.

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58. Factoring Networks by a Statistical Method

Nelson Morgan, Hervé Bourlard

https://doi.org/10.1162/neco.1992.4.6.835 | pp. 835-838

We show that it is possible to factor a multilayered classification network with a large output layer into a number of smaller networks, where the product of the sizes of the output layers equals the size of the original output layer. No assumptions of statistical independence are required.

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59. Maximum Entropy and Learning Theory

Griff L. Bilbro, D.E. Van den Bout

https://doi.org/10.1162/neco.1992.4.6.839 | pp. 839-853

We derive the learning theory recently reported by Tishby, Levin, and Solla (TLS) directly from the principle of maximum entropy instead of statistical mechanics. The theory generally applies to any problem of modeling data. We analyze an elementary example for which we find the predictions consistent with intuition and conventional statistical results and we numerically examine the more realistic problem of training a competitive net to learn a one-dimensional probability density from samples. The TLS theory is useful for predicting average training behavior.

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60. Tight Bounds on Transition to Perfect Generalization in Perceptrons

Yuh■Dauh Lyuu, Igor Rivin

https://doi.org/10.1162/neco.1992.4.6.854 | pp. 854-862

“Sudden” transition to perfect generalization in binary perceptrons is investigated. Building on recent theoretical works of Gardner and Derrida (1989) and Baum and Lyuu (1991), we show the following: for α > α c = 1.44797 …, if α n examples are drawn from the uniform distribution on {+1, –1} n and classified according to a target perceptron w t ∈ {+1, –1} n as positive if w t · x ≥ 0 and negative otherwise, then the expected number of nontarget perceptrons consistent with the examples is 2 –■(√n) ; the same number, however, grows exponentially 2 ■(n) if α ≤ α c . Numerical calculations for n up to 1,000,002 are reported.

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61. Learning Factorial Codes by Predictability Minimization

Jürgen Schmidhuber

https://doi.org/10.1162/neco.1992.4.6.863 | pp. 863-879

I propose a novel general principle for unsupervised learning of distributed nonredundant internal representations of input patterns. The principle is based on two opposing forces. For each representational unit there is an adaptive predictor, which tries to predict the unit from the remaining units. In turn, each unit tries to react to the environment such that it minimizes its predictability. This encourages each unit to filter "abstract concepts" out of the environmental input such that these concepts are statistically independent of those on which the other units focus. I discuss various simple yet potentially powerful implementations of the principle that aim at finding binary factorial codes (Barlow et al. 1989), i.e., codes where the probability of the occurrence of a particular input is simply the product of the probabilities of the corresponding code symbols. Such codes are potentially relevant for (1) segmentation tasks, (2) speeding up supervised learning, and (3) novelty detection. Methods for finding factorial codes automatically implement Occam's razor for finding codes using a minimal number of units. Unlike previous methods the novel principle has a potential for removing not only linear but also nonlinear output redundancy. Illustrative experiments show that algorithms based on the principle of predictability minimization are practically feasible. The final part of this paper describes an entirely local algorithm that has a potential for learning unique representations of extended input sequences.

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62. How to Incorporate New Pattern Pairs without Having to Teach the Previously Acquired Pattern Pairs

H. Wabgaonkar, Allen R. Stubberud

https://doi.org/10.1162/neco.1992.4.6.880 | pp. 880-887

In this paper, we deal with the problem of associative memory synthesis. The particular issue that we wish to explore is the ability to store new input-output pattern pairs without having to modify the path weights corresponding to the already taught pattern pairs. The approach to the solution of this problem is via interpolation carried out in a Reproducing Kernel Hilbert Space setting. An orthogonalization procedure carried out on a properly chosen set of functions leads to the solution of the problem.

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63. Local Learning Algorithms

Léon Bottou, Vladimir Vapnik

https://doi.org/10.1162/neco.1992.4.6.888 | pp. 888-900

Very rarely are training data evenly distributed in the input space. Local learning algorithms attempt to locally adjust the capacity of the training system to the properties of the training set in each area of the input space. The family of local learning algorithms contains known methods, like the k-nearest neighbors method (kNN) or the radial basis function networks (RBF), as well as new algorithms. A single analysis models some aspects of these algorithms. In particular, it suggests that neither kNN or RBF, nor nonlocal classifiers, achieve the best compromise between locality and capacity. A careful control of these parameters in a simple local learning algorithm has provided a performance breakthrough for an optical character recognition problem. Both the error rate and the rejection performance have been significantly improved.

---

64. Object Discrimination Based on Depth-from-Occlusion

Leif H. Finkel, Paul Sajda

https://doi.org/10.1162/neco.1992.4.6.901 | pp. 901-921

We present a model of how objects can be visually discriminated based on the extraction of depth-from-occlusion. Object discrimination requires consideration of both the binding problem and the problem of segmentation. We propose that the visual system binds contours and surfaces by identifying “proto-objects”—compact regions bounded by contours. Proto-objects can then be linked into larger structures. The model is simulated by a system of interconnected neural networks. The networks have biologically motivated architectures and utilize a distributed representation of depth. We present simulations that demonstrate three robust psychophysical properties of the system. The networks are able to stratify multiple occluding objects in a complex scene into separate depth planes. They bind the contours and surfaces of occluded objects (for example, if a tree branch partially occludes the moon, the two "half-moons" are bound into a single object). Finally, the model accounts for human perceptions of illusory contour stimuli.

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65. An Adaptive Lattice Architecture for Dynamic Multilayer Perceptrons

Andrew D. Back, Ah Chung Tsoi

https://doi.org/10.1162/neco.1992.4.6.922 | pp. 922-931

Time-series modeling is a topic of growing interest in neural network research. Various methods have been proposed for extending the nonlinear approximation capabilities to time-series modeling problems. A multilayer perceptron (MLP) with a global-feedforward local-recurrent structure was recently introduced as a

new approach to modeling dynamic systems. The network uses adaptive infinite impulse response (IIR) synapses (it is thus termed an IIR MLP), and was shown to have good modeling performance. One problem with linear IIR filters is that the rate of convergence depends on the covariance matrix of the input data. This extends to the IIR MLP: it learns well for white input signals, but converges more slowly with nonwhite inputs. To solve this problem, the adaptive lattice multilayer perceptron (AL MLP), is introduced. The network structure performs Gram-Schmidt orthogonalization on the input data to each synapse. The method is based on the same principles as the Gram-Schmidt neural net proposed by Orfanidis (1990b), but instead of using a network layer for the orthogonalization, each synapse comprises an adaptive lattice filter. A learning algorithm is derived for the network that minimizes a mean square error criterion. Simulations are presented to show that the network architecture significantly improves the learning rate when correlated input signals are present.

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66. A Model Circuit for Cortical Temporal Low-Pass Filtering

Reinoud Maex, Guy A. Orban

https://doi.org/10.1162/neco.1992.4.6.932 | pp. 932-945

We propose that the low-pass characteristics in the temporal and velocity domain of area 17 cells are generated by the abundant excitatory connections between cortical neurons. We have incorporated this anatomical feature in a model circuit in which simple cells' firing is initiated by geniculocortical excitatory synaptic input, with a short time course, and firing is maintained by feedback corticocortical excitatory synapses, which have a longer time course. The low-pass performance of the model is demonstrated by computing the model simple cells' velocity response curves (VRC) elicited by moving bars, and comparing these to those of its LGN (lateral geniculate nucleus) inputs. For the same parameter set, the VRCs of sustained and transient LGN cells are transformed into VRCs typical of central area 17 and central area 18 cells, respectively.

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67. A "Thermal" Perceptron Learning Rule

Marcus Frean

https://doi.org/10.1162/neco.1992.4.6.946 | pp. 946-957

The thermal perceptron is a simple extension to Rosenblatt's perceptron learning rule for training individual linear threshold units. It finds stable weights for nonseparable problems as well as separable ones. Experiments indicate that if a good initial setting for a temperature parameter, T 0 , has been found, then the thermal perceptron outperforms the Pocket algorithm and methods based on gradient descent. The learning rule stabilizes the weights (learns) over a fixed training period. For separable problems it finds separating weights much more quickly than the usual rules.

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68. Acknowledgment to Reviewers

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69. Index

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